

Curriculum Vitae

Hwan Heo

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RESEARCH INTEREST

Computer Vision
3D Generative AI
Neural Rendering; Radiance Fields & Gaussian Splatting

EXPERIENCE

Lead AI Researcher Feb 2025 - present
NC AI, VARCO3D Team

- Lead technical directions of the 3D generative systems behind VARCO 3D

AI Researcher Mar 2023 - Feb 2025
NCSOFT AI Lab, 3D Generation & Neural Rendering Team

- 3D Asset Generation; 3D Latent Diffusion & Texture Synthesis
- Accelerated NeRF Face Avatar w/ Multi-Camera
- Neural Rendering scene in Graphical Engine

AI Researcher, Intern Jul 2022 - Dec 2022
NAVER AI Lab, Mentor: Ph.D. Jin-Hwa Kim

- NeRF Research Project: Robust Camera Pose Refinement for Multi-Resolution Hash Encoding [ICML '23]

EDUCATION

Master's degree in Computer Science & Engineering Sep 2020 - Feb 2023
Korea University, Advisor: Prof. Hyunwoo J. Kim

Bachelor's degree in Computer Science & Engineering Mar 2015 - Aug 2020
Korea University, Advisor: Prof. HeuiSeok Lim

PUBLICATION

[1] **CaPa: Carve-n-Paint Synthesis for Efficient 4K Textured Mesh Generation**

H. Heo, J. Kim, S. Lee, J. Wi, J. Choi, S. Ahn* [Technical Report] 🔗

[2] **Robust Camera Pose Refinement for Multi-Resolution Hash Encoding**

H. Heo, T. Kim, J. Lee, J. Lee, S. Kim, H. J. Kim*, JH. Kim* [ICML 2023] 🔗

[3] **Semantic-aware Occlusion Filtering Neural Radiance Fields in the Wild**

J. Lee, I. Kim, H. Heo, H. J. Kim* [preprint] 🔗

[4] **Consistency Learning via Decoding Path Augmentation for Transformers in Human Object Interaction Detection**

*J. Park, S. Lee, **H. Heo**, H. Choi, H. J. Kim** [CVPR 2022] 

[5] **Temporally Sparse Adversarial Perturbations on Videos**

***H. Heo**, D. Ko, J. Lee, Y. Hong*, H. J. Kim** [IEEE Access 2021] 

PROJECT

VARCO3D; NC AI's 3D Asset Generation Service 
NC AI

Jul 2024 -

AI-based 3D Rendering in Game Engine 
NCSOFT Research

May 2023 - Dec 2023

Object Detection for Diagnosis of Infectious Diseases

Korea University and Korea University Guro Hospital

Aug 2020 - Jul 2022

**COMPUTER
SKILLS**

Languages & Software: Python, PyTorch, CUDA

PROFESSIONAL *Invited Talks*

ACTIVITIES

- An Introduction to 3D Generative AI - KAIST, Dec 2025
- Spatial 3D AI Technology and its Applications - Ministry of Land, Infrastructure and Transport in Korea, Sep 2025
- Radiance Fields and Spatial 3D AI for Digital Twin - National Geographic Information Institute in Korea, Aug 2025
- Rasterization vs. Ray Tracing with 3D Gaussian Primitives - NCSOFT developer seminar, Sep 2024
- Joint Optimization of Camera Pose and Accelerated Radiance Fields - NAVER Cloud, Mar 2023

Technical Assistant

- Object Detection Tutorial in KCCV 2022
- Object Detection, Adversarial Attack and GAN Tutorial in Korea University-LG Electronics Winter AI Workshop 2022